# Project Learning from data

# Writing a statistical report

**Zoë Pieters**

**2024-2025**



DSI DATA SCIENCE INSTITUTE

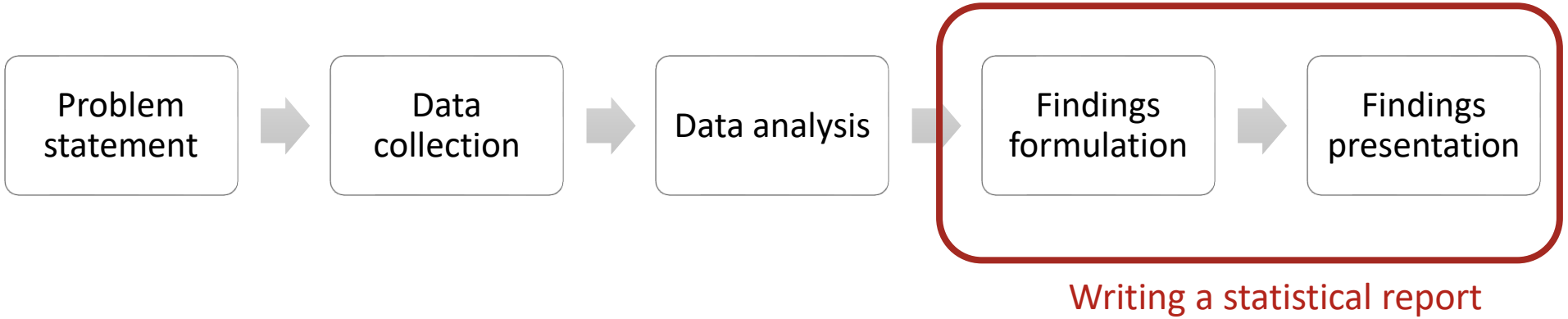
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# Project learning from data

- Practical
- Prepares you for your future career

- In general, the following process is followed:



# Content of the seminar

1. **Before writing a statistical report**
2. **Writing a statistical report**
3. **Different elements of a statistical report**
4. **Guidelines on reporting**
5. **Do’s and Don’ts**
6. **Reflection on the example**

# Who will read the report?

Consider the intended audience:

- Basic understanding: understands basic statistical measures and charts
- Intermediate understanding: understands the more commonly used statistical concepts and terms
- Advanced understanding: understands sophisticated statistical terminology

For the scope of project learning from data, you are writing a report for an audience with advanced understanding, i.e. other statisticians.

# Criteria for a written report

- Content:
  - Clear goal
  - Precise
  - Concise

Report must be self-contained

- Use of references, in an appropriate way, to credit the sources used.
- Presentation:
  - Tidy
  - Legible
  - Where needed/appropriate, tables and figures can be used.

# Example data

A randomized clinical trial was conducted on 20 patients, 10 of whom received Drug A and the other 10 received Drug B. After taking the medication for 4 weeks, the blood pressure was measured of each of the 20 patients. The goal is to evaluate whether there is a difference in lowering blood pressure between Drug A and B.

# Example 1

**Research question:**

Is there a difference in lowering blood pressure between Drug A and B?

**Results:**

```
> t.test(drugA, drugB, alternative="two.sided", paired=FALSE,
+        var.equal= TRUE,
+        conf.level=0.95)

        Two Sample t-test

data:  drugA and drugB
t = -0.49924, df = 18, p-value = 0.6237
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -7.532365  4.639900
sample estimates:
mean of x mean of y
 17.73648  19.18272
```

Positive aspects:

- Research question is well defined

Negative aspects:

- Content:
  - Not self-contained
  - No explanation nor interpretation is provided
- Presentation:
  - Crucial elements of a statistical report are lacking
  - R output is copied directly into the report

# Trial comparing blood pressure reduction between Drug A and B

**Background**

A randomized clinical trial was conducted on 20 patients, 10 of whom received Drug A and the other 10 received Drug B. After taking the medication for 4 weeks, the blood pressure was measured of each of the 20 patients. The goal is to evaluate whether there is a difference in lowering blood pressure between Drug A and B.

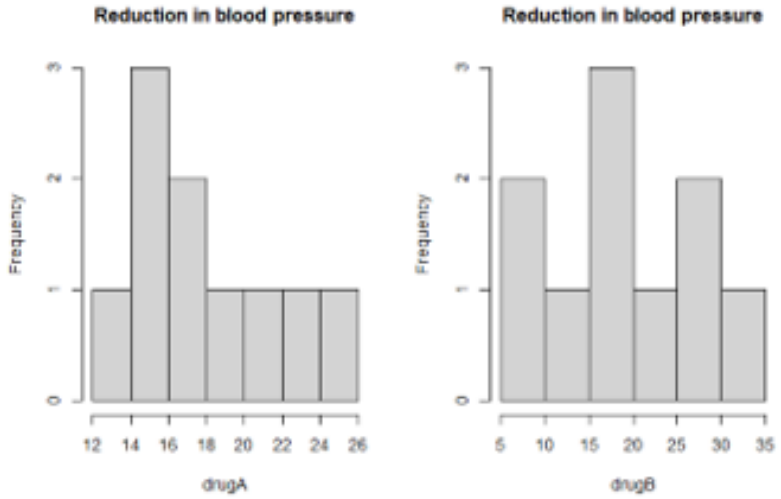
**Research question**

Is there a difference in lowering blood pressure between Drug A and B?

**Statistical method**

A two-sample t-test is used to test for a difference in blood pressure reduction between Drug A and Drug B. The significance level is set to 0.05.

**Results**




We do not reject the hypothesis of equal average reduction in blood pressure for the two groups and conclude that there is insufficient evidence reduction in blood pressure differs between Drug A and Drug B. The software shows a p-value of 0.6237.

# Example 2

Positive aspects:

- Different sections are used to structure the report.
- Results from the test are interpreted

Negative aspects:

- Assumptions underlying the test are not described
- The figures have different ranges on the x-axes and no caption is provided.

# What elements/sections should a report contain?

- Abstract
- Background
- Data management
- Statistical methods
- Statistical results
- Discussion
- References
- Appendix

SAMPL guidelines

SAMPL = Statistical Analyses and Methods in Published Literature

# Abstract

- Optional
  - Leave out: short report with few results
  - Include: large project with many results
- Summary of what's been done and to show the main results

# Introduction

- Limited
- Descripe the scope and set-up of the study
- Demonstrate the understanding of the study

  → **What** is it about? **Why** is the study being conducted? **How** is the study being conducted?

# Data management – Data cleaning

- Continuous variables: check for plausible values, e.g. blood pressure reduction of 1200 mmHg?
- Categorical variables: consistency in notation
  - M= Male
  - F= Female
  - V=?
- General: variables might consist of a mix of numbers and characters

| PatientID | reduction |
|---|---|
| 2032001 | 15 |
| 2032002 | 8 |
| … | … |
| 2032059 | **1200** |

## Different elements of a statistical report

Reporting statistical methods

- Preliminary analyses
- Primary analyses

Reporting statistical results

- Descriptive statistics
- Hypothesis tests
- Linear models

# Reporting statistical methods

## Preliminary analyses

Identify any statistical procedures used to modify raw data before analysis

- Transforming continuous measurements to make distributions closer to a normal distribution
- Creating ratios
- Collapsing continuous data into categories
- …

# Reporting statistical methods and results using the example

**Research question:**

Is there a difference in lowering blood pressure between Drug A and B?

**Results:**

```
> t.test(drugA, drugB, alternative="two.sided", paired=FALSE,
+         var.equal= TRUE,
+         conf.level=0.95)

        Two Sample t-test

data:  drugA and drugB
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alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -7.532365  4.639900
sample estimates:
mean of x mean of y
 17.73648  19.18272
```

# Reporting statistical methods

## Primary analyses

- Describe the purpose of the analysis (AIM)
- Identify the variables used in the analysis and mention how you will report them (e.g. mean and SD, frequency and %,…)
- Describe the methods used to analyze the data
- Mention whether tests were one- or two-tailed
- Verify that the assumptions are met
- Indicate whether adjustments for multiple comparisons were made
- Mention how outliers were treated
- Report the significance level
- Name the statistical package or program used

**Our Example**

- Evaluate whether drug A and drug B result in a different reduction in blood pressure
- Drug and blood pressure reduction (mmHg); mean and standard deviation (SD)
- Two-sample t-test
- Two-tailed
- Independence, normality, homogeneity
- No adjustments
- No outliers were removed
- Significance level of 5%
- Statistical software R, version 4.4.1

# Reporting statistical results

## Descriptive statistics

- Summarize normally distributed data: mean (SD)
- Summarize non-normally distributed data: median, interquartile range
- Report numerators and denominators for all %
- Report total sample size and group sample size

**Our Example**

- Drug A:  17.74 (4.18); Drug B: 19.18 (8.15)
- Drug A:  16.92 (14.57-20.93); Drug B: 18.97 (14.43-25.69)
- Total sample size=20; 10 patients received drug A and 10 patients received drug B.

# Reporting statistical results

## Hypothesis tests

- State the hypothesis/purpose
- State the test
- Confirm the assumptions
- Report the alpha level
- Report measure of precision (95%CI)
- Report exact p values
- Report adjustments for multiple comparisons

**Our Example**

- Evaluate whether drug A and drug B result in a different reduction in blood pressure
- Two-tailed two-sample t-test
- All assumptions were met
- Significance level 5%
- -1.45; 95% CI: -7.53 – 4.64
- P value = 0.6237
- Not applicable

# Reporting statistical results

## Linear models

- State the purpose
- Confirm the assumptions
- If relevant, report how outliers were treated
- If model selection was performed, describe the process
- If multiple independent variables are present, were interactions assessed
- Report the coefficients of the independent variables and the p values
- Provide an assessment of the goodness-of-fit

# Reporting statistical methods and results



There is not a one-size-fits-all report.



Depending on the methods, need for extra elements/topics.



Crucial to report all necessary information such that the analyses can be reproduced.

# Discussion

- Summarise the key results
- Interpret the results considering the objectives, limitations, ..
- Discuss the limitaions of the statistical analysis, the study design,…

# References

- Credit sources that were used!
- Be consistent
- Different citation styles exist, such as author-data or numeric citation styles.
- Suggestion:
  - Use a citation manager (endnote, mendeley,..) that retreives the citation from article databases
  - Automatically integrate the citations in your document and bibliography

# References

## Numeric citation style

In-text Citation

However, one of the major problems with only accounting for uncertainty in a deterministic way is that the estimated cost-effectiveness, and its associated threshold values, can be incorrect in the case of a nonlinear relationship between the input parameters and the model's outcomes.[1]

Probabilistic sensitivity analysis (PSA) can overcome this limitation.[2,3] PSA accounts for the plausible values

Bibliography

Research article

8 Bilcke J, Antillón M, Pieters Z, Kuylen E, Abboud L, Neuzil KM, et al. Cost-effectiveness of routine and campaign use of typhoid Vi-conjugate vaccine in Gavi-eligible countries: a modelling study. The Lancet Infectious Diseases. 2019;19(7):728-739.

Book

6 Briggs A, Claxton K, Sculpher M. Decision modelling for health economic evaluation. Oxford University Press; 2006.

Webpage

11 Blumsack S. Decision Analysis Without Probabilities: Threshold and Sensitivity Analysis. Available from: https://www.e-education.psu.edu/eme801/node/577. Accessed 17 January 2019.

# References

## Author-date citation style

In-text Citation

The past few decades, the global prevalence of obesity has increased tremendously (Leddy et al., 2008; Finkelstein et al., 2005; Visscher TLS and Visscher TL and Seidell JC, 2001). According to the World Health Organization (WHO), the prevalence has already doubled from 1980 to 2008

Bibliography

Similar to numeric citation style, the difference is that the articles are listed in alphabetical order rather than order of appearance.

# Appendix

- Contains tables, figures, codes
- It supports the reader's understainding of the work, but is not fundamental for understanding it.
- For instance:

*Clinical Infectious Diseases*

REVIEW ARTICLE

IDSA Infectious Diseases Society of America hivma hiv medicine association OXFORD

## Case Fatality Rate of Enteric Fever in Endemic Countries: A Systematic Review and Meta-analysis

**Zoë Pieters,[1,2] Neil J. Saad,[3] Marina Antillón,[3] Virginia E. Pitzer,[3,a] and Joke Bilcke[1,a]**

[1]Vaccine & Infectious Disease Institute, University of Antwerp, Wilrijk, and [2]Department of Mathematics and Statistics, Hasselt University, Diepenbeek, Belgi
of Microbial Diseases, Yale School of Public Health, New Haven, Connecticut

sion and exclusion criteria (Table 1). We searched MEDLINE, PubMed Central, Embase, and Web of Science for eligible articles using terms related to "mortality," "typhoid fever," and "paratyphoid fever" on 11 January 2017, without language restrictions. A detailed overview of the search strategy is provided in the Supplementary Material (Supplement 1): Supplementary Figures 1–3. Titles and abstracts were screened

# Appendix

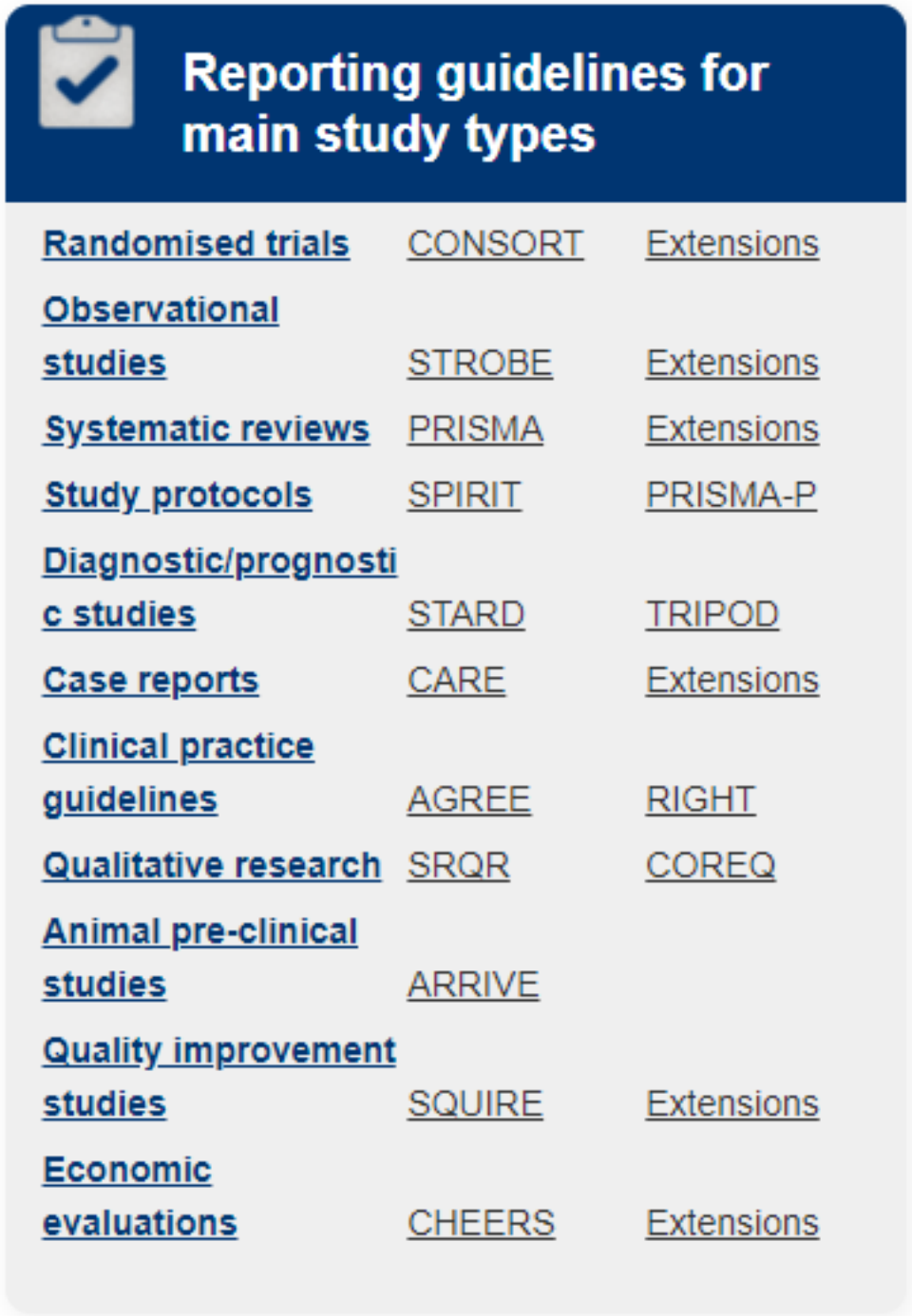
- Contains tables, figures, codes
- It supports the reader's understanding of the work,
  but is not fundamental for understanding it.
- For instance:

## A Computationally Efficient Method for Probabilistic Parameter Threshold Analysis for Health Economic Evaluations

Zoë Pieters, Mark Strong, Virginia E. Pitzer, Philippe Beutels, and Joke Bilcke

**Algorithm 2** Regression-Based Scheme for Estimating Threshold Value $\theta_i^*$ for Parameter $\theta_i$

1 Sample $K$ times from the joint distribution of all parameters $p(\boldsymbol{\theta})$.
2 Order sampled values of $\boldsymbol{\theta}$ with respect to $\theta_i$ such that $\theta_i^{(1)} < \theta_i^{(2)} < \ldots < \theta_i^{(K-1)} < \theta_i^{(K)}$.
3 **for** $k = 1$ **to** $K$ **do**
4 Evaluate the incremental net benefit function for each $d$ and store $INB_d(\boldsymbol{\theta}^{(k)})$. This is the standard "PSA" sample.
5 **end**
6 **for** $d = 1$ **to** $D$ **do**
7 Regress $\mathrm{INB}_d(\boldsymbol{\theta}^{(1,\ldots,K)})$ on $\theta_i^{(1,\ldots,K)}$ using a GAM (R code available in Appendix A, available online).
8 Compute the regression fitted values, $\widehat{\mathrm{INB}}_d^{(1,\ldots,K)}$. These are estimates of the conditional expected values $\mathrm{E}_{\boldsymbol{\theta}_{-i}|\theta_i}[\mathrm{INB}_d(\theta_i^{(1,\ldots,K)}, \boldsymbol{\theta}_{-i})]$.
9 **end**
10 **for** $k = 1$ **to** $K$ **do**
11 Set $\widehat{\mathrm{INB}}_d^{(k)} = 0$ for the "baseline" decision option $d = 0$.
12 Determine the policy option with the highest expected INB given $\theta_i^{(k)}$, $d^{(k)} = \arg\max_d \widehat{\mathrm{INB}}_d^{(k)}$.
13 **end**
14 Determine any value(s), $k^*$, such that $d^{(k^*)} \neq d^{(k^*+1)}$ (see note).
15 Each $k^*$ will define a threshold value $\theta_i^*$ that lies in the interval $\theta_i^{(k^*)} < \theta_i^* < \theta_i^{(k^*+1)}$ (A function, written in R,[18] is available in Appendix A, available online).




Guidelines on reporting

# Equator Network

**Enhancing the QUAlity and Transparency Of health Research**

MISSION: achieve accurate, complete, and transparent reporting of all health research studies to support research reproducibility and usefulness

These guidelines are tools to improve quality of reporting for specific studies and are not ment to evaluate the quality of the reseach nor ar they a guide fo designing or conducting studies.

REFERENCE: https://www.equator-network.org/reporting-guidelines/

# When to include figures and tables?

- Group information together
- Summarize information
- Report information in an orderly fashion (without describing it in the text)
- Specifically for figures/graphs: communicate patterns, relationships or trends

Always use labels and captions when including figures, graphs or tables in a report!

# Guidelines for Tables and Figures

- Number them in the order they appear in the text
- Reference (and discuss) **all** tables and figures that were included in the text

- Tables:
  - The caption is placed above the table and includes a table number and descriptive title
  - Rows and columns in the table are labeled

- Figures:
  - The caption is placed below the figure and includes a figure number and descriptive title

# Tables

The MEANS Procedure

| Variable | N | Mean | Std Dev | Minimum | Maximum |
|---|---|---|---|---|---|
| Sepal_Length | 8 | 6.4875000 | 0.4051014 | 5.9000000 | 7.0000000 |
| Sepal_Width | 8 | 3.2000000 | 0.1069045 | 3.1000000 | 3.4000000 |

- What is reported?
- Is all the information listed in the output needed in the report?
- Caption?
- Decimals?

Do not copy-paste output from the statistical software straight into the report!

Make sure all tables are consistent with respect to the format.

# Tables

Covariance Parameter Estimates

| Cov Parm | Subject | Estimate | Standard Error | Z Value | Pr Z |
|---|---|---|---|---|---|
| UN(1,1) | Subject | 244.98 | 52.4690 | 4.67 | <.0001 |
| UN(2,1) | Subject | 218.98 | 49.5772 | 4.42 | <.0001 |
| UN(2,2) | Subject | 241.93 | 52.1518 | 4.64 | <.0001 |
| UN(3,1) | Subject | 2.5596 | 0.9464 | 2.70 | 0.0068 |
| UN(3,2) | Subject | 3.0978 | 0.9835 | 3.15 | 0.0016 |
| UN(3,3) | Subject | 0.1266 | 0.02893 | 4.38 | <.0001 |
| UN(4,1) | Subject | 2.0556 | 0.7391 | 2.78 | 0.0054 |
| UN(4,2) | Subject | 2.2284 | 0.7411 | 3.01 | 0.0026 |
| UN(4,3) | Subject | 0.07852 | 0.01977 | 3.97 | <.0001 |
| UN(4,4) | Subject | 0.07207 | 0.01733 | 4.16 | <.0001 |
| Residual | | 12.6588 | 0.2946 | 42.97 | <.0001 |

**Table 3.** Covariance parameter estimates and standard errors of the statistical model for the integrated approach.

| | Estimate | Standard Error |
|---|---|---|
| Variance (Int session 1) | 244.98 | 52.46 |
| Covariance (Int session 1, Int session 2) | 218.98 | 49.58 |
| Variance (Int session 2) | 241.93 | 52.15 |
| Covariance (Int session 1, Slope session 1) | 2.56 | 0.95 |
| Covariance (Int session 2, Slope session 1) | 3.1 | 0.98 |
| Variance (Slope session 1) | 0.13 | 0.03 |
| Covariance (Int session 1, Slope session 2) | 2.06 | 0.74 |
| Covariance (Int session 1, Slope session 2) | 2.23 | 0.74 |
| Covariance (Slope session 1, Slope session 2) | 0.08 | 0.02 |
| Variance (Slope session 2) | 0.07 | 0.02 |
| Residual | 12.66 | 0.29 |

Abbreviations: *Int* intercept.

Schoutteten et al. (2024)

Do's and Don'ts

# Tables

**Table 3. CFR According to Stratification Factors[a]**

| Categories | Studies, No. | $I^2$, % | CFR (95% CI), % |
|---|---|---|---|
| World Bank income level | | | |
| Low | 29 | 96.72 | 2.28 (1.19–4.31) |
| Lower-middle | 12 | 35.50 | 3.17 (2.29–4.38) |
| Upper-middle | 2 | 6.66 | 5.21 (0–99.84) |
| WHO region | | | |
| African | 16 | 94.20 | 3.82 (1.97–7.26) |
| Eastern Mediterranean | 2 | 0 | 3.17 (.13–45.28) |
| Americas | 3 | 0 | 3.45 (2.00–5.89) |
| South-East Asia | 15 | 90.92 | 2.27 (.91–5.55) |
| Western Pacific | 8 | 40.72 | 1.26 (.66–2.36) |
| Detection method | | | |
| Serology and cultures | 24 | 92.96 | 3.22 (2.08–4.97) |
| Cultures only | 19 | 94.48 | 1.63 (.63–4.12) |
| Inclusion of HIV-infected individuals | | | |
| Yes | 5 | 84.91 | 4.20 (1.02–15.74) |
| No | | 95.03 | 2.32 (1.49–3.61) |
| Serovar | | | |
| *S.* Typhi and *S.* Paratyphi | 12 | 85.11 | 3.08 (1.55–6.04) |
| *S.* Typhi only | 28 | 93.26 | 2.36 (1.37–4.04) |
| Not specified | 2 | 0 | 10.6 (4.82–21.74) |
| Countries with multiple estimates | | | |
| Bangladesh | 2 | 0 | 1.60 (0–99.29) |
| India | 7 | 82.57 | 4.52 (1.91–10.32) |
| Indonesia | 3 | 0 | 6.74 (2.06–19.90) |
| Nigeria | 4 | 97.81 | 4.28 (.40–33.34) |
| Senegal | 3 | 75.45 | 6.20 (.29–60.37) |
| South Africa | 2 | 62.41 | 2.16 (0–100) |
| Vietnam | 4 | 0 | 0.53 (.13–2.16) |

Abbreviations: CI, confidence interval; CFR, case fatality rate; HIV, human immunodeficiency virus; *S.* Paratyphi and *S.* Typhi, *Salmonella enterica* serovars Paratyphi and Typhi; WHO, World Health Organization.

[a]More information is provided in Supplement 5.

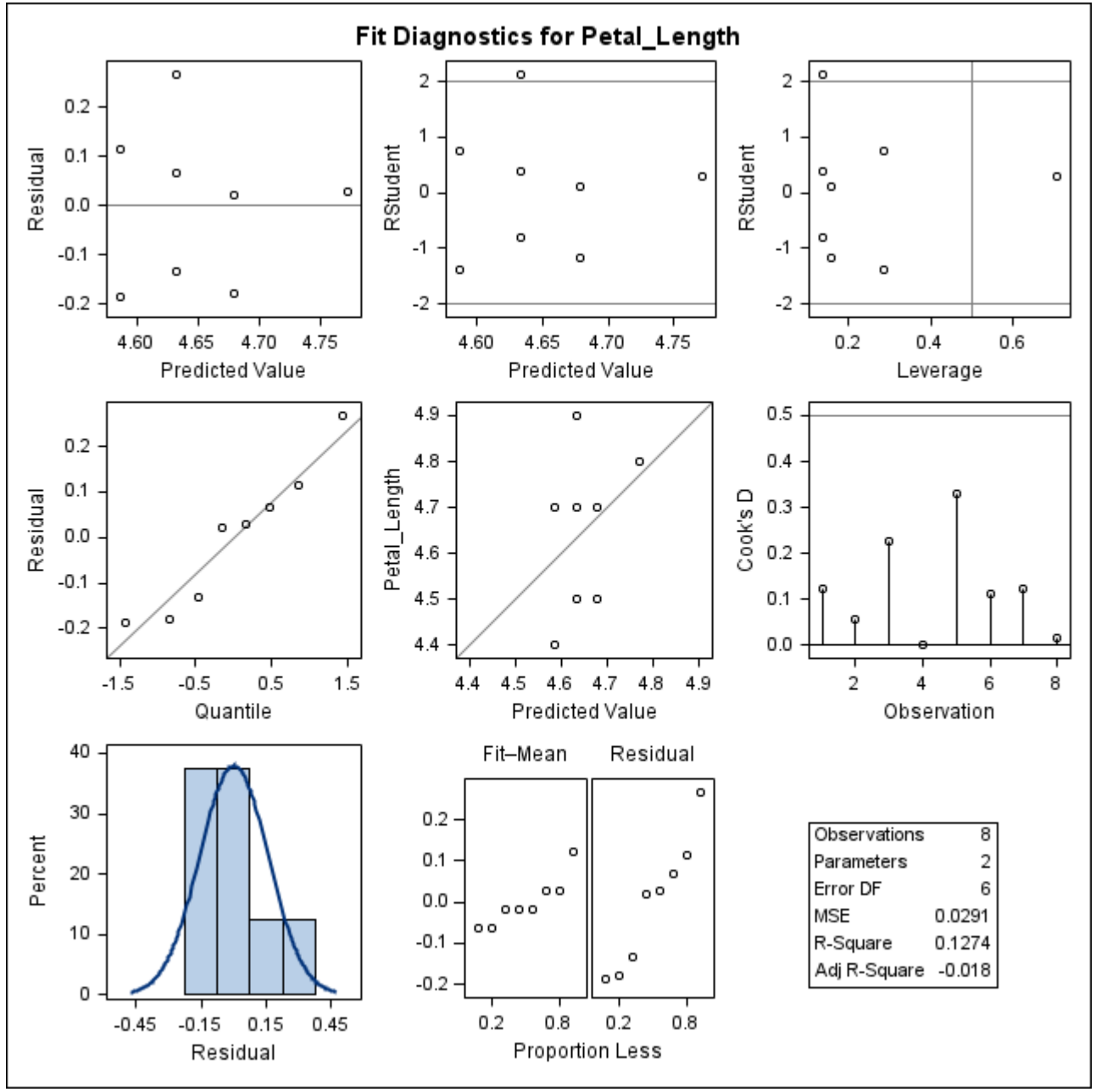
Do's and Don'ts

# Figures

- What is reported in the different plots?
- Is all the information listed in the figure needed/discussed in the report?
- Caption?

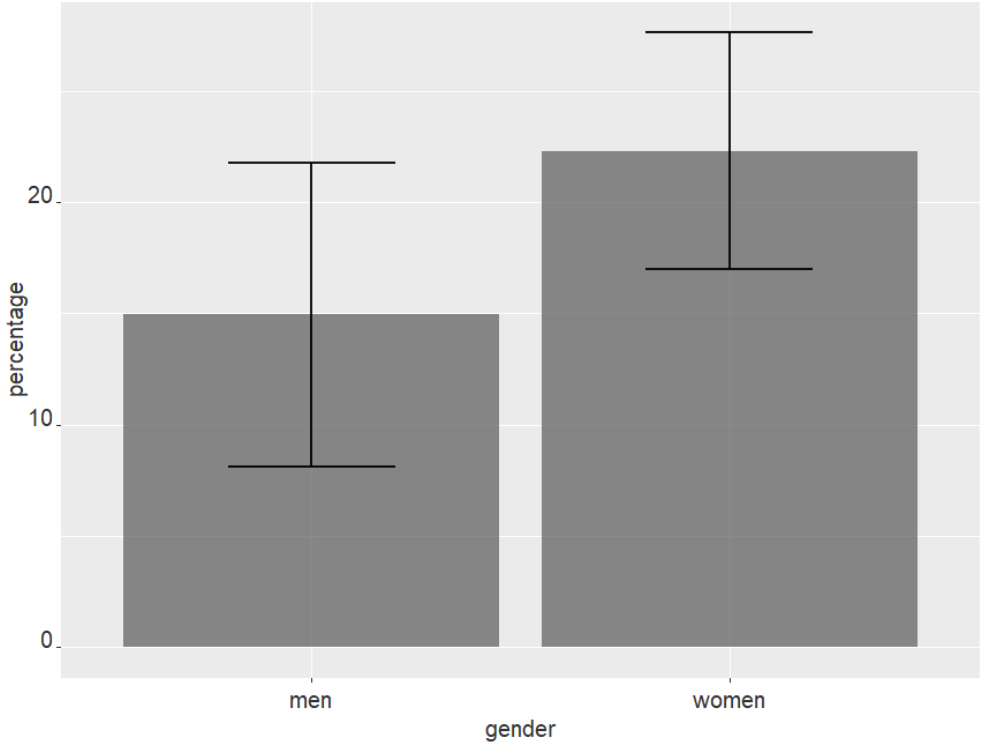
Do not copy-paste output from the statistical software straight into the report!

Only include the necessary figures.

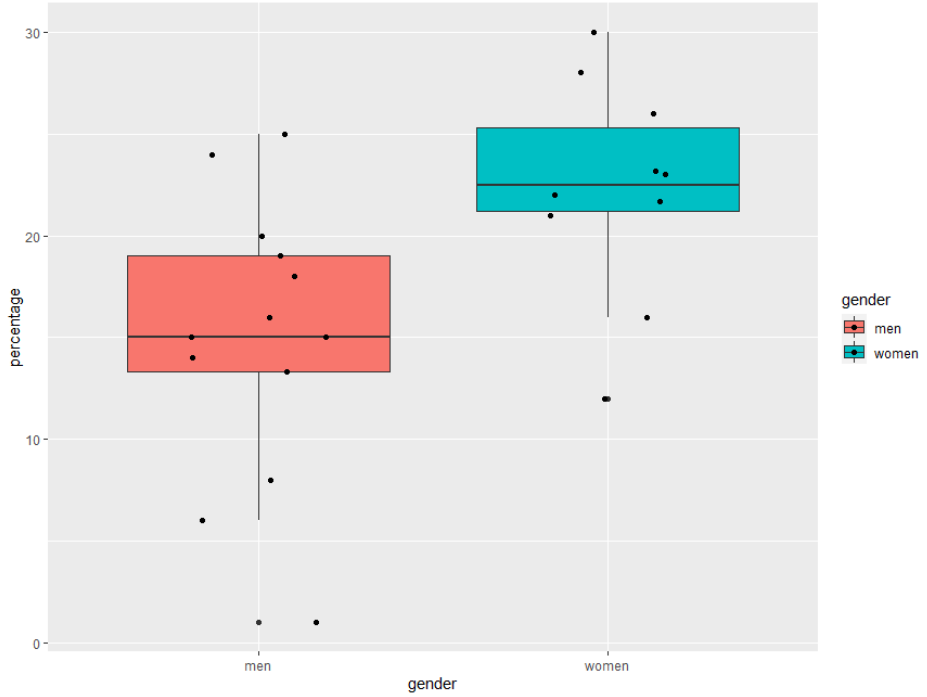


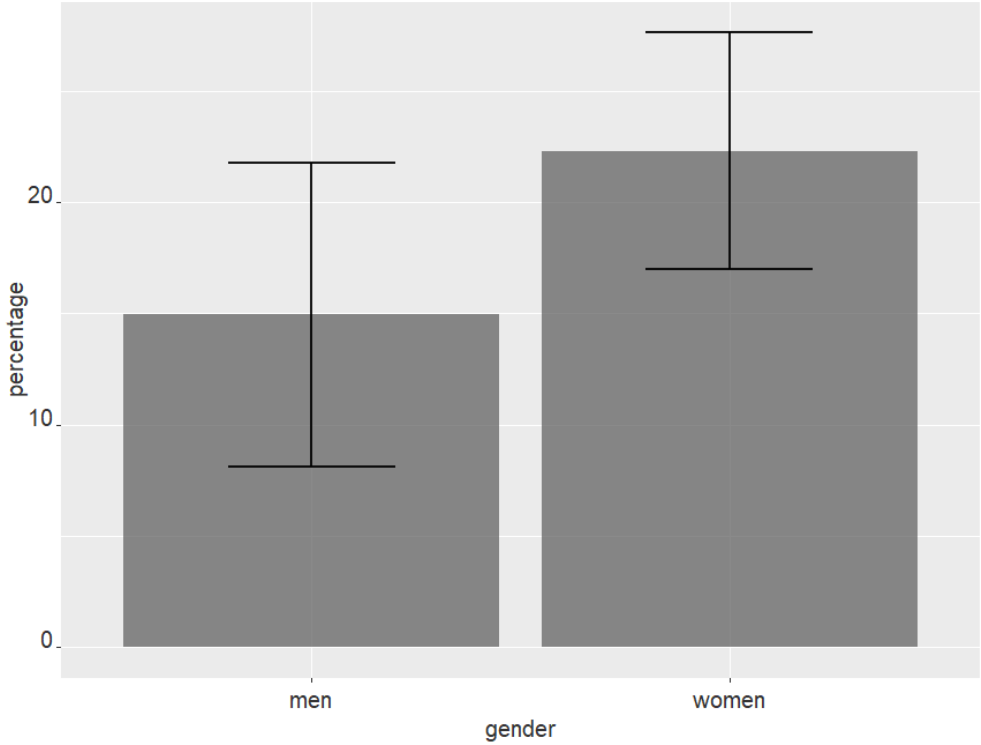
# Figures

- High ink-to-information ratio
- Bars do not visualize the variation well
- Do the bars represent the SD, SEM, 95%CI?



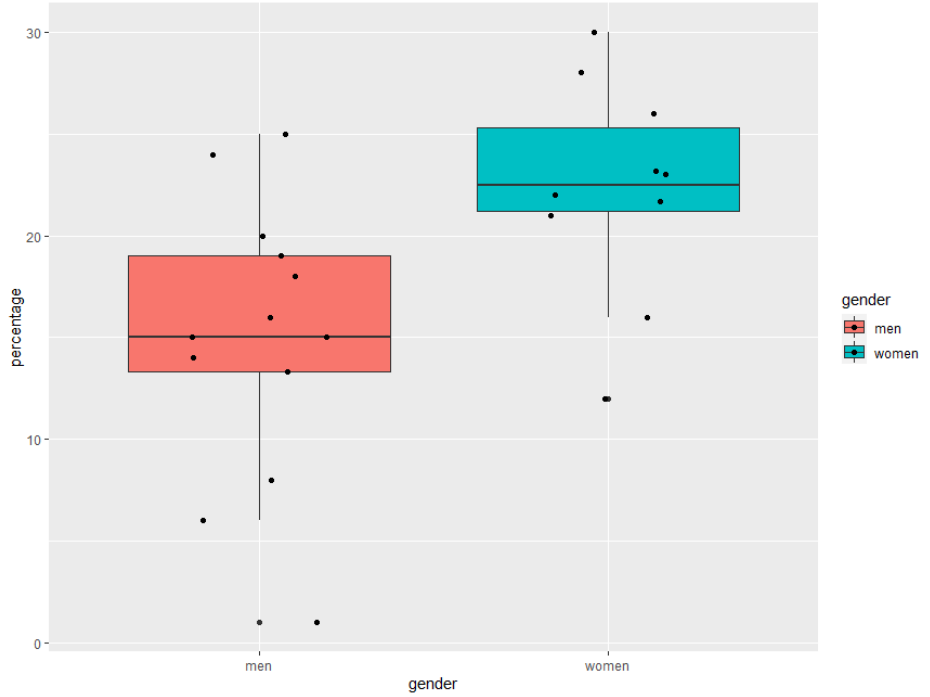
# Figures


30
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gender
men
women

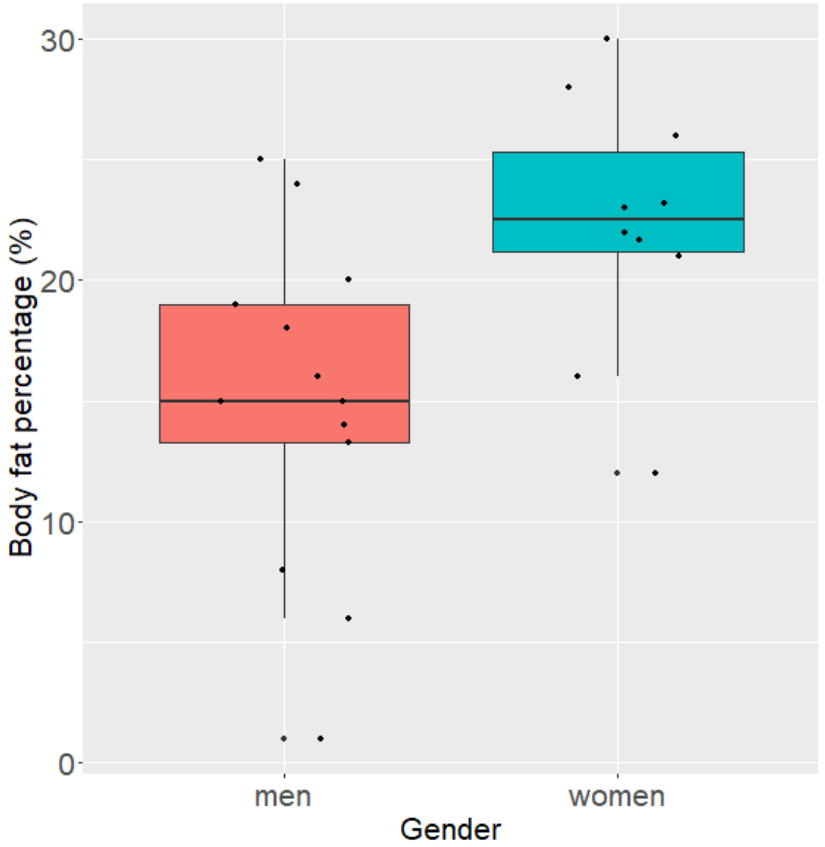

20
10
0
percentage
men
women
gender

# Figures

- Labels need to be readable
- Add descriptive labesl to the y- and x-axis
- Add a numbered caption that explains the graph



# Figures



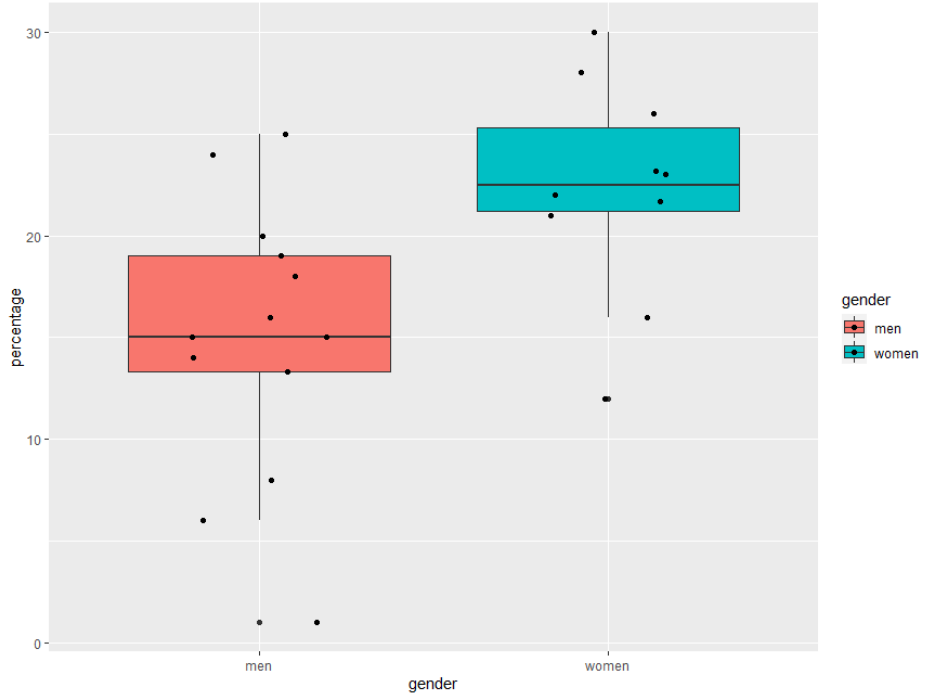




***Figure 1.*** Body fat percentage according to gender. A boxplot as well as observed body fat percentages (dots) are displayed according to gender.

# Trial comparing blood pressure reduction between Drug A and B

**Background**

A randomized clinical trial was conducted on 20 patients, 10 of whom received Drug A and the other 10 received Drug B. After taking the medication for 4 weeks, the blood pressure was measured of each of the 20 patients. The goal is to evaluate whether there is a difference in lowering blood pressure between Drug A and B.

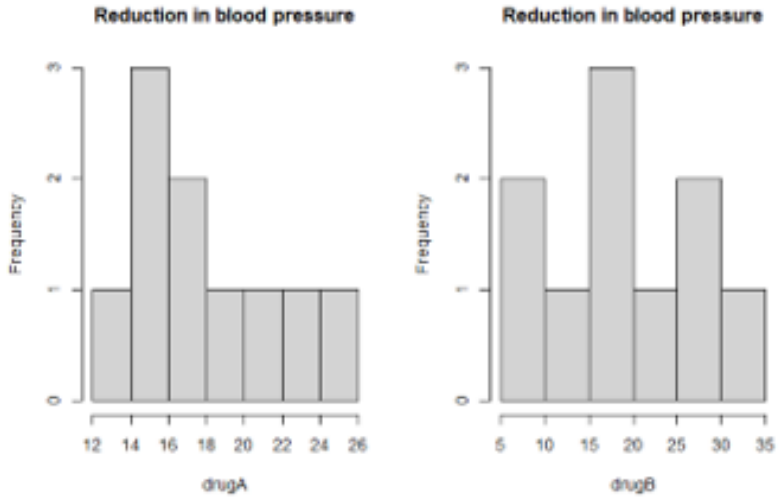
**Research question**

Is there a difference in lowering blood pressure between Drug A and B?

**Statistical method**

A two-sample t-test is used to test for a difference in blood pressure reduction between Drug A and Drug B. The significance level is set to 0.05.

**Results**




We do not reject the hypothesis of equal average reduction in blood pressure for the two groups and conclude that there is insufficient evidence reduction in blood pressure differs between Drug A and Drug B. The software shows a p-value of 0.6237.

Reflection on an example

# Example 2

What would you improve?

# References

Content:


- Kenett R. and Thyregod P. (2006) Aspects of statistical consulting not taught by academia, Statistica Neerlandica, 60(3):396-411
- Lang T, Altman D. (2016) Statistical analyses and methods in published literature: The SAMPL guidelines
- Reporting guidelines: https://www.equator-network.org/reporting-guidelines/prisma/


Examples:


- Pieters Z, Saad NJ, Antillón M, Pitzer VE, Bilcke J. Case Fatality Rate of Enteric Fever in Endemic Countries: A Systematic Review and Meta-analysis. Clin Infect Dis. 2018 Aug 1;67(4):628-638. doi: 10.1093/cid/ciy190.
- Pieters Z, Strong M, Pitzer VE, Beutels P, Bilcke J. A Computationally Efficient Method for Probabilistic Parameter Threshold Analysis for Health Economic Evaluations. Med Decis Making. 2020 Jul;40(5):669-679. doi: 10.1177/0272989X20937253.
- Schoutteten MK, Lindeboom L, De Cannière H, Pieters Z, Bruckers L, Brys ADH, van der Heijden P, De Moor B, Peeters J, Van Hoof C, Groenendaal W, Kooman JP, Vandervoort PM. The Feasibility of Semi-Continuous and Multi-Frequency Thoracic Bioimpedance Measurements by a Wearable Device during Fluid Changes in Hemodialysis Patients. Sensors (Basel). 2024 Mar 15;24(6):1890. doi: 10.3390/s24061890.

# Questions?